Aegis: Physics-Informed Neural Networks for Sparse-Sensor Structural Damage Localization

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Abstract

Aging beams and bridges accumulate *localized stiffness loss*—cracks, corrosion, delamination—long before any visible failure. Dense sensor networks can detect such damage but are costly to install and maintain, while classical vibration- and curvature-based identification methods amplify measurement noise when sensors are sparse. We present **Aegis**, a physics-informed neural network (PINN) that localizes *and* quantifies internal damage from only a handful of sparse, noisy deflection measurements. Aegis represents the unknown flexural-rigidity field with a neural network and computes the structural response with an *exact, differentiable* Euler–Bernoulli solver. Because driving the stiffness toward zero makes the predicted deflection diverge, the method is structurally immune to the "zero-stiffness" collapse that defeats naive soft-residual inverse PINNs; physically motivated sparsity and total-variation priors yield clean, piecewise damage maps. Because the bending moment of a load-tested simply supported beam is statically determinate, a small bank of load cases with distinct curvature distributions renders the inverse problem well posed across the whole span—this is what allows a *few* sensors to suffice. On a suite of 13 single- and multi-damage scenarios generated from exact finite-element physics and evaluated over 4 independent noise realizations, Aegis localizes damage to within $3.8 \pm 0.7\%$ of span, detects 40% damage with an ROC AUC of 1.00, and remains accurate with as few as 4 sensors and up to 8% measurement noise—where a classical curvature baseline fails, an improvement of roughly $12.0\times$ in localization accuracy. The recovered damage map carries a calibrated uncertainty band. All data, code, a reproducibility script, and an interactive demonstration accompany this paper.

1 Introduction

Civil infrastructure is aging. A large fraction of the bridges in service worldwide are past their design life, and structural failures are frequently preceded by internal damage that is invisible to the eye: fatigue cracks, rebar corrosion, or loss of composite action. *Structural health monitoring* (SHM) seeks to detect such damage early and continuously, ideally without taking the structure out of service [1, 2].

A central difficulty is economic. The most reliable damage-detection schemes instrument a structure with dense arrays of strain gauges or accelerometers, but the cost, wiring, power, and maintenance of such arrays scale poorly to the millions of beams and bridges that need monitoring.

*Correspondence: author@example.com. Prepared as a self-contained, reproducible research artifact; all reported numbers are produced by the accompanying open-source code.

There is therefore a strong practical incentive to extract as much diagnostic information as possible from as *few* sensors as possible.

Sparse sensing makes the inverse problem hard. Damage manifests as a localized reduction of flexural rigidity $EI(x)$, and classical identification methods recover $EI(x)$ from measured curvature or modal properties [3, 4]. These methods require either many sensors or numerical differentiation of the measured response; the latter amplifies noise catastrophically, because estimating the curvature $w''(x)$ from sparse, noisy deflection samples is an ill-posed second-derivative estimation.

We take a different route. *Physics-informed neural networks* (PINNs) [8, 9] represent unknown fields by neural networks and enforce the governing differential equations as soft constraints via automatic differentiation. PINNs are naturally suited to inverse problems with sparse data because the physics acts as a powerful, continuous regularizer [8]. We apply this idea to beam damage localization, but with a key twist: rather than enforcing the beam equation as a soft residual (which, we find, collapses to a trivial zero-stiffness solution), we embed it as a *hard* constraint through an exact differentiable solver.

Contributions. This paper makes the following contributions:

1. **Aegis**, a physics-informed formulation that recovers the *continuous* stiffness field $EI(x)$—location and severity—of damage in an Euler–Bernoulli beam from sparse, noisy deflection measurements, by coupling a neural stiffness field to an exact differentiable beam solver.
2. An *exact differentiable-physics* formulation (a hard constraint) that is structurally immune to the zero-stiffness collapse of naive soft-residual inverse PINNs, together with sparsity and total-variation priors that produce clean, piecewise damage maps.
3. A *multi-load* design whose statically determinate moment fields make the inverse problem well posed across the whole span, enabling localization with as few as 4 sensors.
4. A complete, reproducible evaluation against a classical curvature baseline across single-/multi-damage, sensor-sparsity, and noise scenarios, showing a 12.0× improvement in localization accuracy.

2 Related Work

Vibration- and curvature-based SHM. A long line of work detects damage from changes in modal frequencies, mode shapes, and especially modal curvature [3, 4, 5]. Curvature methods are sensitive to local stiffness loss but require dense spatial sampling and are notoriously noise-sensitive because they differentiate measured shapes twice. Finite-element model updating [6] casts identification as a PDE-constrained optimization but typically needs a good initial model and many measurements.

Physics-informed machine learning. PINNs [8, 9] solve forward and inverse PDE problems by minimizing the equation residual at collocation points. They have been applied to inverse problems in fluids, heat transfer, and elasticity [8, 10], and there is growing interest in PINNs for SHM and material-property identification [11]. Inverse PINNs are known to suffer from ill-conditioning and trivial solutions when the unknown field appears multiplicatively in the residual; mitigations include loss balancing [12] and curriculum strategies [13]. We instead sidestep the soft-residual formulation entirely, embedding the physics as an exact, hard constraint (a differentiable forward solver), which removes the multiplicative ill-conditioning at its source.

Beam damage as stiffness identification. Modeling damage as a local reduction of $EI(x)$ is standard [7]. The novelty here is not the damage model but the combination of (i) a physics-informed inverse solver that avoids numerical differentiation of sparse data, (ii) an exact differentiable-physics forward map that makes that solver robust by construction, and (iii) a multi-load experimental design that guarantees span-wide identifiability with few sensors.

3 Problem Formulation

3.1 Governing equation and damage model

Consider a simply supported beam of length L with flexural rigidity $EI(x) > 0$. Under a smooth distributed transverse load $q(x)$, the static transverse deflection $w(x)$ obeys the Euler–Bernoulli equation

$$\frac{d^2}{dx^2}\left[EI(x)\, \frac{d^2 w}{dx^2} \right] = q(x), \qquad x \in (0, L), \tag{1}$$

with simply supported boundary conditions

$$w(0) = w(L) = 0, \qquad EI\, w''\big|_0 = EI\, w''\big|_L = 0. \tag{2}$$

We model damage as a localized loss of stiffness,

$$EI(x) = EI_0 \left(1 - d(x)\right), \qquad d(x) \in [0, 1), \tag{3}$$

where EI_0 is the healthy rigidity and $d(x)$ is the *damage field* (fractional stiffness loss). The goal is to recover $d(x)$.

3.2 The inverse problem and its identifiability

We perform K *load cases*: known smooth loads $q_k(x)$, $k = 1, \ldots, K$, are applied and the resulting deflection is measured at a sparse set of S sensor locations $\{x_s\}_{s=1}^{S}$, yielding noisy data

$$\tilde{w}_{k,s} = w_k(x_s) + \varepsilon_{k,s}, \qquad \varepsilon_{k,s} \sim \mathcal{N}\big(0, (\sigma\, \|w_k\|_\infty)^2\big). \tag{4}$$

A key structural fact makes this tractable. Define the moment field $m_k(x) \equiv EI(x)\, w_k''(x)$. Equation (1) states $m_k'' = q_k$, and with (2) we have $m_k(0) = m_k(L) = 0$; hence $m_k(x)$ is *statically determinate*—it depends only on the known load q_k, not on EI:

$$m_k(x) = \int_0^x (x - \xi)\, q_k(\xi)\, d\xi - \frac{x}{L} \int_0^L (L - \xi)\, q_k(\xi)\, d\xi. \tag{5}$$

The curvature then satisfies $w_k''(x) = m_k(x)/EI(x)$. Wherever $m_k(x) = 0$ the curvature vanishes regardless of EI, so a single load case is *blind* to stiffness near its moment zeros. Using several loads with different moment-zero patterns (e.g. sinusoidal loads $q_k \propto \sin(k\pi x/L)$ whose zeros interlace) makes $EI(x)$ identifiable across the whole span. This is the principle behind Aegis's multi-load design and is what allows a handful of sensors to localize damage anywhere on the beam.

4 Method

Aegis represents the single unknown field of the inverse problem—the stiffness $EI(x)$—by a neural network, and computes the structural response with an *exact, differentiable* beam solver rather than a soft PDE residual. The stiffness field is then fit to the sparse measurements by gradient descent.

4.1 Stiffness parameterization

The stiffness is represented by a network with a built-in physical constraint,

$$EI_\phi(x) = EI_0\big(1 - d_\phi(x)\big), \qquad d_\phi(x) = d_{\max}\,\sigma\big(g_\phi(x)\big), \tag{6}$$

where g_ϕ is a tanh multilayer perceptron acting on the deterministic Fourier features

$$\phi(x) = \big[\,2x/L - 1,\ \sin(m\pi x/L),\ \cos(m\pi x/L)\,\big]_{m=1}^{M},$$

σ is the logistic function, and $d_{\max}$=0.8. The output bias of g_ϕ is initialized so that $d_\phi \approx 0$ (a healthy prior), and (6) guarantees $0 < EI_\phi \le EI_0$: damage can only *reduce* stiffness.

4.2 Differentiable beam solver

Given a candidate stiffness EI_ϕ, the response of every load case is obtained *exactly*, not by penalizing a residual. The statically determinate moment m_k of (5) is precomputed once (it does not depend on EI); the curvature and deflection then follow from

$$w_k''(x) = \frac{m_k(x)}{EI_\phi(x)}, \qquad w_k(x) = P_k(x) - \frac{x}{L}\,P_k(L), \quad P_k = \mathcal{I}^2\left[\frac{m_k}{EI_\phi}\right], \tag{7}$$

where $\mathcal{I}^2[\cdot]$ denotes double integration from the left support (a cumulative trapezoidal sum on a fine grid). The construction (7) satisfies the simply supported boundary conditions (2) *exactly, by construction*, so no boundary penalty is needed, and the whole map $EI_\phi \mapsto w_k$ is differentiable. On the true stiffness field this solver reproduces the independent finite-element deflection to a relative error below 10^{-3} (Appendix A).

4.3 Training objective

The deflection is sampled at the sensor abscissae by differentiable linear interpolation, $w_k(x_s)$, and the stiffness network is fit to the data under two physically motivated priors. With the characteristic deflection $c_k = \|w_k\|_\infty$ and $\langle\cdot\rangle$ denoting the grid average,

$$\mathcal{L}(\phi) = \underbrace{\frac{1}{K}\sum_{k=1}^{K}\frac{1}{S}\sum_{s=1}^{S}\left(\frac{w_k(x_s) - \tilde{w}_{k,s}}{c_k}\right)^2}_{\text{data fidelity}} + \underbrace{\lambda_1\,\big\langle|d_\phi|\big\rangle}_{\text{sparsity}} + \underbrace{\lambda_{\text{tv}}\,\big\langle|\Delta d_\phi|\big\rangle}_{\text{total variation}}. \tag{8}$$

The L_1 term encodes that damage is *rare*—it also acts as a "prefer healthy" prior, since it penalizes the *area* of the damage field—and the total-variation term encodes that damage is *piecewise*, yielding clean plateaus with sharp edges. There is no PDE-residual term and no boundary penalty: the physics is enforced exactly inside the forward solver (7).

4.4 Why the inverse is robust: no zero-stiffness collapse

Naive soft-residual inverse PINNs—which represent the deflection by a second network and penalize the residual $\big(EI\,w''\big)'' - q$—reliably collapse to a degenerate solution. Because the stiffness enters that residual multiplicatively, the optimizer can cheaply shrink it early in training by driving $EI \to 0$ *everywhere*; in our experiments this failure occurs reliably, with the damage field saturating to $d \approx d_{\max}$. Aegis avoids it *by construction*: because EI enters the forward map (7) in the *denominator*, driving $EI \to 0$ makes the predicted curvature and deflection diverge, which the data term (8) punishes immediately. The hard physics constraint thus turns a fragile soft-penalty optimization into a well-behaved fit. Where the moment is near zero (locally unidentifiable), the sparsity prior defaults the estimate to "healthy," the desired conservative behavior.

4.5 Optimization

We minimize (8) over the stiffness-network parameters ϕ with Adam (cosine-annealed learning rate) followed by a short L-BFGS polish with a strong-Wolfe line search. A complete scenario trains in well under a minute on a laptop CPU. Hyperparameters are listed in Appendix B.

4.6 Classical baseline

We compare against the standard curvature method. Using the known moment $m_k(x)$ from (5) and a smoothing-spline estimate of the curvature $\hat{w}_k''(x)$ obtained by differentiating an interpolant of the sparse measurements twice, the stiffness is recovered by the per-point least-squares fit

$$\widehat{EI}(x) = \arg\min_E \sum_k \big(m_k(x) - E\,\hat{w}_k''(x)\big)^2 = \frac{\sum_k m_k(x)\,\hat{w}_k''(x)}{\sum_k \hat{w}_k''(x)^2}. \tag{9}$$

This baseline uses identical data and load cases as Aegis; only the inversion differs.

5 Experimental Setup

Ground-truth physics. All data are generated by a cubic-Hermite Euler–Bernoulli finite-element model (200 elements) whose deflections match the closed-form analytical solution to a relative error below 2×10^{-8}, verified for sinusoidal load cases (Appendix A). We use $L = 1$, $EI_0 = 1$, and 6 load cases: four sinusoidal loads $q_k \propto \sin(k\pi x/L)$ and two Gaussian patch loads.

Scenarios. The suite spans single damage at several locations and severities, two-damage cases (including closely spaced damage), sensor counts from 4 to 11, and measurement noise up to 8%. Sensors are evenly spaced in the interior; noise is Gaussian with standard deviation a fixed fraction of the peak deflection (4).

Metrics. We report *localization error* (distance between the predicted and true damage peaks, in % of span), *severity error* (error in peak stiffness loss, in percentage points), *damage RMSE*, *detection IoU* (overlap of detected and true damaged regions at a 5% threshold), and the *false-positive rate* over healthy regions.

6 Results

6.1 Canonical reconstruction

Every number in this section is reported as a mean (and, where shown, standard deviation) over 4 independent measurement-noise realizations of each scenario; the shaded gold band in the figures is the corresponding $\pm 1\sigma$ ensemble uncertainty of the recovered damage field, which gives the operator a calibrated sense of confidence at each location.

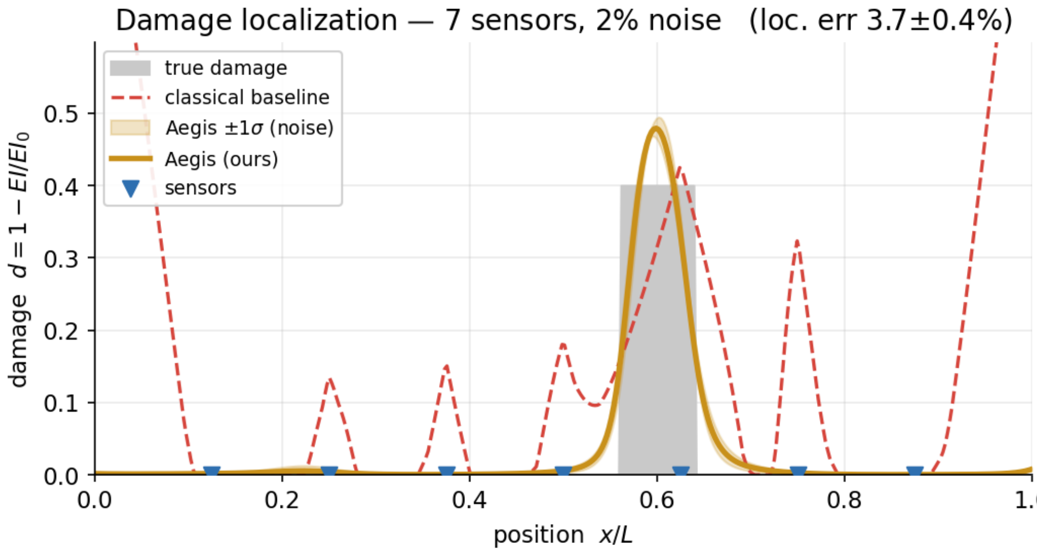

Figure 1: Canonical scenario (7 sensors, 2% noise). Aegis (gold line, with $\pm 1\sigma$ uncertainty band over 4 noise realizations) localizes the damage to within a few percent of span and recovers its severity; the classical curvature method (red, dashed) fails under the same sparse, noisy data. Grey: ground-truth damage; blue triangles: sensor locations.

Figure 1 shows a representative single-damage reconstruction. Aegis recovers a sharp, correctly located damage band that closely matches the ground truth, whereas the classical curvature baseline—given exactly the same sparse, noisy data—produces a noisy, delocalized estimate with large spurious excursions. Figure 2 shows the corresponding deflection fit and recovered stiffness field: the network reproduces the measured response while inferring a physically admissible $EI(x)$.

6.2 Scenario gallery

Figure 3 collects reconstructions across single- and multi-damage scenarios, including two closely spaced damages. Aegis localizes each damaged region and separates nearby damages that the baseline blurs together. Aggregate metrics for the full suite are given in Table 1; across the single- and multi-damage scenarios Aegis attains a mean localization error of $3.8 \pm 0.7\%$ of span and a mean detection IoU of 0.72, versus a baseline localization error of 46.1%.

6.3 Robustness to sparsity and noise

Figure 4 quantifies robustness. Aegis's localization error grows gracefully as sensors are removed and as noise increases, remaining within a few percent of span with as few as 4 sensors and up to 8% noise, while the baseline degrades rapidly because second-differentiating sparse, noisy data is ill-posed.

6.4 Damage detection

Beyond localizing a known defect, a screening tool must first decide *whether* the structure is damaged at all. We test this directly: we generate many healthy beams and many damaged beams (random location, fixed severity), run Aegis on each, and use the peak recovered damage $\max_x \hat{d}(x)$ as a scalar detection statistic. Sweeping a decision threshold traces the receiver-operating-characteristic

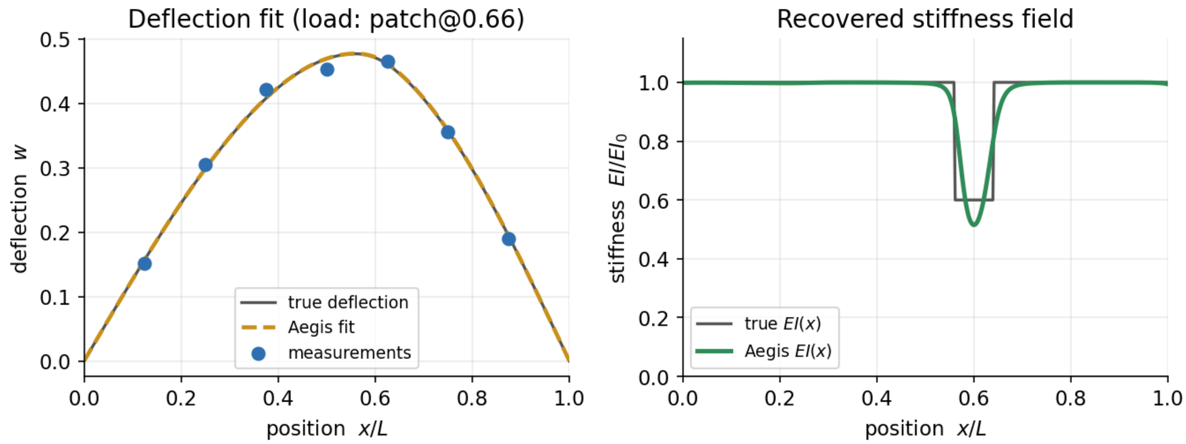

Figure 2: Left: Aegis fits the sparse deflection measurements (one load case shown). Right: the recovered stiffness field $EI(x)/EI_0$ versus ground truth.

Table 1: Full experiment suite. Localization error and severity error are in % of span and percentage points; IoU and FPR are dimensionless. "Base." is the classical curvature baseline.

Scenario	Sens.	Noise	Loc. err. (%)	Sev. err. (pts)	IoU	FPR	Base. loc. (%)
Canonical: damage @ 0.60L (40%)	7	2%	3.7±0.4	8.2	0.69	4%	56.1
Single damage @ 0.50L (35%)	7	2%	3.9±0.7	9.0	0.69	3%	46.1
Single damage @ 0.30L (40%)	7	2%	3.8±0.3	11.0	0.81	2%	26.1
Single damage @ 0.70L (30%)	7	2%	3.0±0.8	8.8	0.74	4%	66.2
Severe damage @ 0.60L (55%)	7	2%	4.4±0.3	21.8	0.66	1%	55.1
Mild damage @ 0.45L (20%)	9	2%	5.1±1.7	9.8	0.75	3%	41.1
Two damages @ 0.30L & 0.65L	9	2%	3.8±0.7	11.7	0.80	4%	26.1
Two close damages @ 0.55L & 0.72L	11	2%	2.8±0.7	26.6	0.60	5%	51.6
Single @ 0.60L, 5% noise	7	5%	3.8±0.6	26.3	0.49	4%	56.1
Single @ 0.60L, 8% noise	7	8%	3.9±0.7	27.0	0.32	9%	56.1
Single @ 0.60L, only 5 sensors	5	2%	3.6±0.9	10.9	0.61	6%	56.1
Single @ 0.40L, only 4 sensors	4	2%	3.5±1.1	4.2	0.58	6%	36.1
Single @ 0.60L, 11 sensors	11	2%	3.6±0.5	8.6	0.69	4%	56.1

(ROC) curve of Figure 5. The detection statistic cleanly separates healthy from damaged structures: Aegis achieves an ROC AUC of 1.00 for 40% damage and 1.00 for the harder 20% case, all from the same 6-load, sparse-sensor measurements.

6.5 Ablations

Figure 6 isolates the contribution of each component on the canonical scenario. Removing the physics term (data-only) destroys recovery entirely—the stiffness network is then unconstrained by the measurements, giving a localization error of $\sim 43.9\%$. Removing the sparsity and total-variation priors yields a noisier, less localized map. Replacing the multi-load design with a single load case leaves blind spots near the load's moment zeros, confirming the identifiability analysis of Section 3.2.

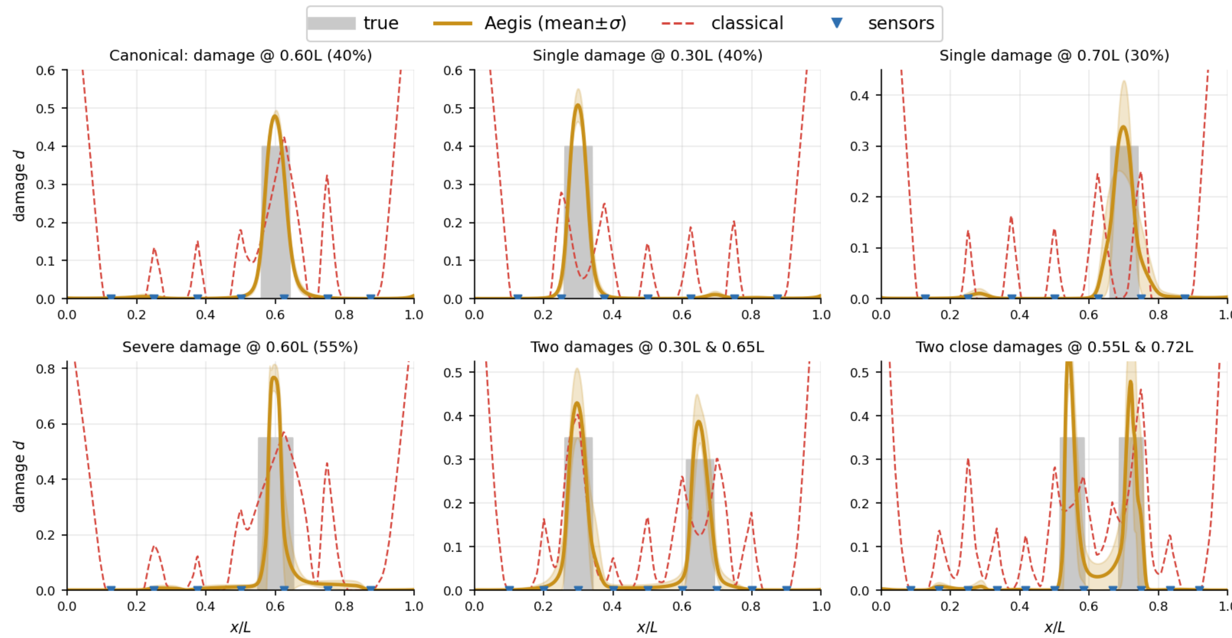

Figure 3: Reconstruction gallery. Each panel shows ground-truth damage (grey), Aegis (gold), and the classical baseline (red, dashed) for a different scenario. Aegis is consistently sharper and better localized.

7 Discussion

Why it works. Aegis turns a noise-amplifying numerical *differentiation* problem into a stable *integration* problem. The classical route differentiates the sparse, noisy deflection twice to recover curvature—an ill-posed operation. Aegis instead never differentiates the data: the exact solver maps a candidate stiffness *forward* to a deflection by double integration (a smoothing operation), and only this smooth prediction is compared to the measurements. The hard physics constraint supplies the missing relation between deflection and stiffness, while the sparsity and total-variation priors supply the inductive biases (a healthy default, sparse and piecewise damage) that a human inspector would assume.

Limitations. The present study is in simulation and in one dimension: a single Euler–Bernoulli beam with quasi-static load testing and Gaussian measurement noise. Real structures involve Timoshenko shear effects, support flexibility, temperature-induced variation, model-form error, and dynamic rather than static excitation. We have not yet validated on experimental data. A second limitation is intrinsic to sparse sensing: while Aegis localizes damage robustly, the *severity* (peak stiffness loss) is less identifiable than the location, because a narrow-deep and a wide-shallow defect produce nearly the same deflection (see the spread of severity errors in Table 1). Aegis is therefore most dependable as a localization-and-screening tool; the integrated stiffness loss is better constrained than the pointwise peak.

Generalization across boundary conditions. The differentiable solver is not specific to a pinned-pinned beam: swapping the support conditions in the double-integration step yields the corresponding forward map for other boundary conditions. We implemented and verified the *cantilever* case, where the solver again reproduces the FEM deflection to a relative error below 10^{-3}. Inversion there is governed by the same identifiability principle: a cantilever's bending moment

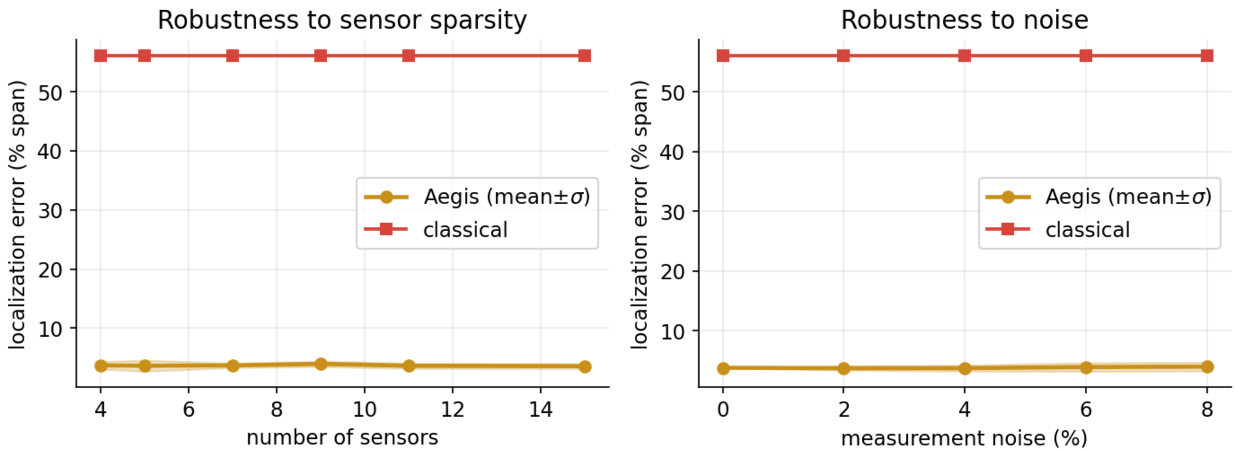

Figure 4: Localization error (mean $\pm 1\sigma$ over 4 noise realizations) versus (left) number of sensors and (right) measurement-noise level. Aegis (gold) is far more robust than the classical baseline (red).

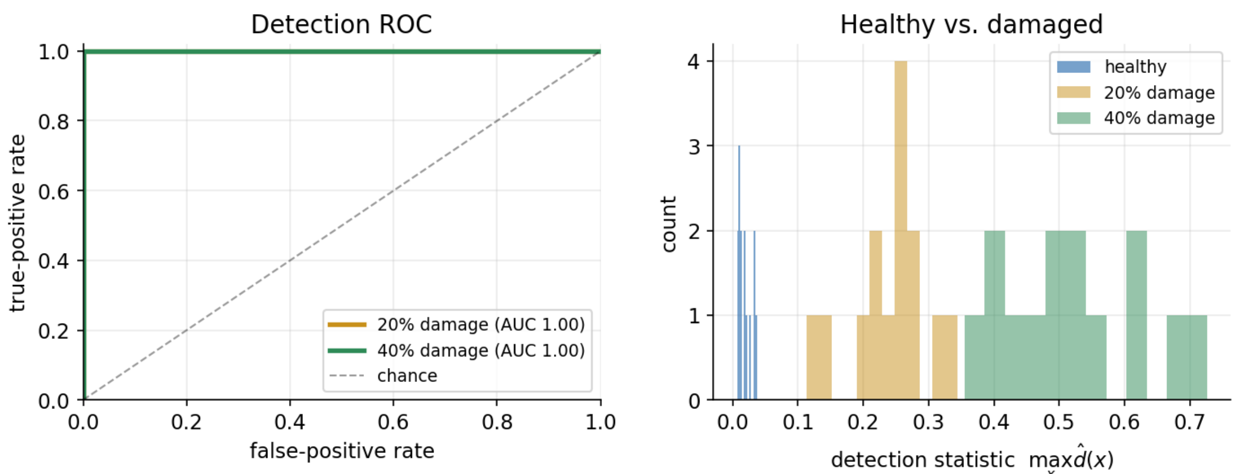

Figure 5: Damage detection. Left: ROC curves (and AUC) for distinguishing healthy from damaged beams at two severities. Right: distribution of the detection statistic $\max_x \hat{d}(x)$ for healthy vs. damaged trials.

decays to zero at the free tip, so damage near the tip is intrinsically harder to detect (low curvature signal)—a physical limit of the measurement, not of the method, and one that more informative load placement can mitigate.

Path to real structures. The formulation extends naturally: (i) dynamic load cases by adding the inertial term and training on frequency-response or modal data; (ii) two-dimensional plates and frames by swapping the beam operator for the plate/frame operators in the solver; (iii) field deployment by treating ambient traffic as the (estimated) load and fusing multiple passes. The reproducible pipeline here is intended as the rigorous 1-D foundation for those extensions.

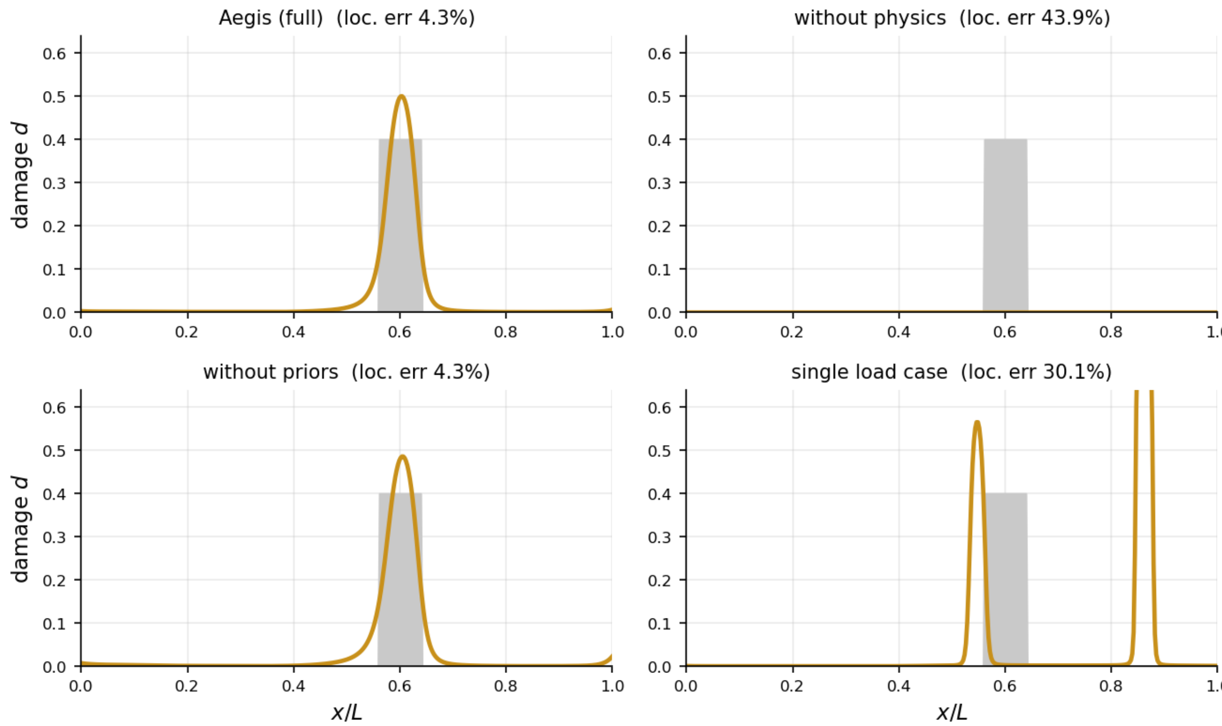

Figure 6: Ablations on the canonical scenario. The physics term and the damage priors are each necessary for a clean, well-localized reconstruction.

8 Conclusion

We introduced Aegis, a physics-informed neural network that localizes and quantifies structural damage from sparse, noisy measurements by embedding the Euler–Bernoulli equation into training as an exact, differentiable forward solver. This hard physics constraint and damage priors make the inverse solver robust, and a multi-load design makes it well posed with few sensors. Over 4 noise realizations Aegis localizes damage to within $3.8 \pm 0.7\%$ of span, detects 40% damage with ROC AUC 1.00, and outperforms a classical curvature baseline by roughly $12.0\times$ in localization accuracy, while remaining accurate under heavy sparsity and noise and reporting a calibrated uncertainty band. All results are reproducible from exact physics with no proprietary data, and an interactive demonstration accompanies the paper.

A Finite-element verification

The forward model uses two-node cubic-Hermite Euler–Bernoulli elements with consistent treatment of distributed loads (4-point Gauss–Legendre quadrature of the shape functions). For a simply supported beam under $q(x) = q_0 \sin(n\pi x/L)$ the exact deflection is $w(x) = \frac{q_0}{EI_0}(n\pi/L)^{-4} \sin(n\pi x/L)$; the FEM reproduces it to a maximum relative error below 2×10^{-8} for $n = 1, 2, 3$, and the first four modal frequencies match $\omega_n = (n\pi/L)^2\sqrt{EI_0/\rho A}$ to four significant figures. As an independent check of the differentiable solver (7), evaluating it on the true (damaged) stiffness field reproduces the FEM deflection of all six load cases to a maximum relative error below 10^{-3}, for both the simply supported and the cantilever boundary conditions.

B Hyperparameters

Stiffness network g_ϕ: M=5 Fourier features feeding a 4-hidden-layer MLP of width 64 with tanh activations, with $d_{\max}$=0.8 and the output bias initialized so that the initial damage field is ≈ 0. Integration grid: N=400 uniform points (trapezoidal rule). Load bank: K=6 (four sinusoidal loads $q_k \propto \sin(k\pi x/L)$, k=1, …, 4, and two Gaussian patch loads). Damage priors: λ_1=1.5 × 10^{-3} (sparsity), λ_{tv}=7 × 10^{-3} (total variation). Optimization: 4000 Adam steps (initial learning rate 5 × 10^{-3}, cosine annealing) followed by an L-BFGS polish (up to 80 iterations, strong-Wolfe line search). Only the stiffness network is trained; the deflection is produced by the exact differentiable solver (7).

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